

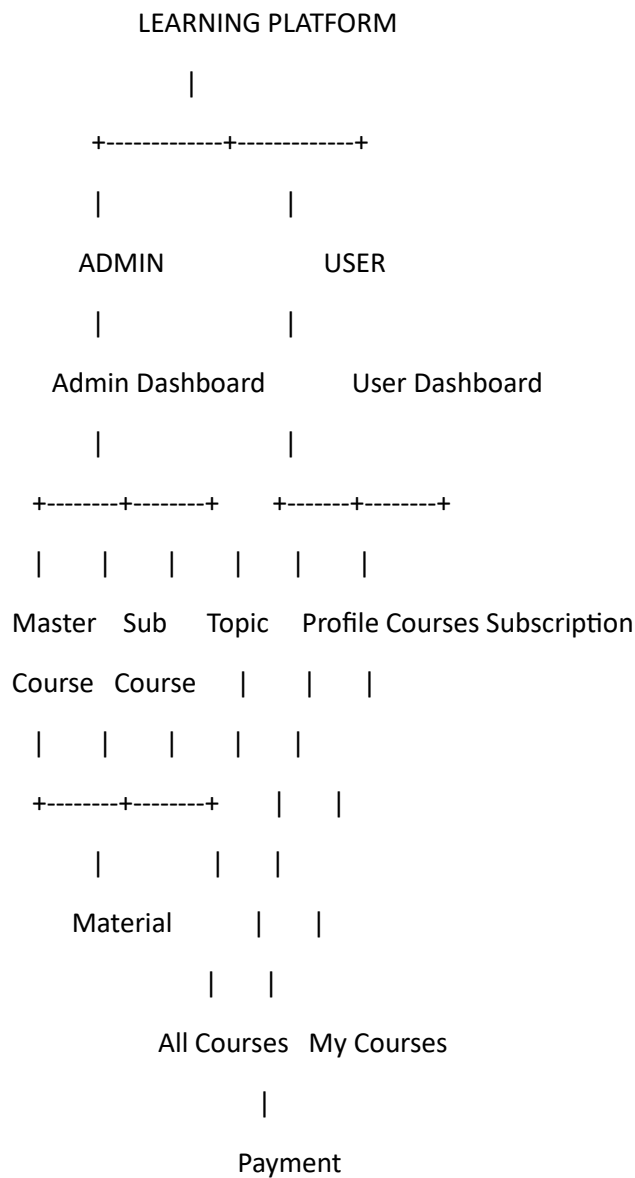
Learning Platform — Project Overview

We are building a **Udemy-like Learning Platform** using:

- **ASP.NET Core MVC**
- **Entity Framework Core**
- **LINQ**
- **SQL Server**
- **Bootstrap**
- **Razorpay**

The application has **two roles: Admin and User**.

1. Overall Project Flow



|
Razorpay

The main course hierarchy is:

Master Course

|
+---- Sub Course

|
+---- Topic

|
+---- Material

2. Admin Side

Admin will have:

Admin Dashboard

Courses

├— Master Course
├— Sub Course
└— Topic

Materials

Subscriptions

Users

Logout

Every Admin management page will follow the same pattern:

GridView

↓

[Add]



Bootstrap Modal



Enter Information



[Save]



Database



Grid Refresh

For Edit:

Grid



[Edit]



Same Bootstrap Modal



Update

3. Master Course

Admin can create and manage master courses.

Grid

ID | Course Name | Price | Status | Action

1 | ASP.NET | 999 | Active | Edit Delete

2 | Java | 799 | Active | Edit Delete

[+ Add]

Add Modal

Add Master Course

Course Name

[_____]

Description

[_____]

Price

[_____]

Status

[Active ▼]

[Cancel] [Save]

4. Sub Course

Sub Course belongs to a Master Course.

Master Course

|

+-- MVC

+-- Web API

+-- Entity Framework

Grid

ID	Master Course	Sub Course	Amount	Status	Action
----	---------------	------------	--------	--------	--------

1	ASP.NET Core	MVC	499	Active	Edit Delete
---	--------------	-----	-----	--------	-------------

Add Modal

Master Course

[ASP.NET Core ▼]

Sub Course Name

[_____]

Amount

[_____]

Status

[Active ▼]

[Save]

5. Topic

Topic belongs to a Sub Course.

Master Course



Sub Course



Topic

Example:

ASP.NET Core



MVC



Controllers

Add Topic Modal

Master Course

[ASP.NET Core ▼]

Sub Course

[MVC ▼]

Topic Name

[_____]

Video URL

[_____]

Status

[Active ▼]

[Save]

6. Material

Material belongs to a Topic.

Admin can add:

- Video
- Assignment/File
- MCQ

Add Material Modal

Master Course

[ASP.NET Core ▼]

Sub Course

[MVC ▼]

Topic

[Controllers ▼]

Material Type

[Video ▼]

Video URL

[_____]

Assignment

[Choose File]

MCQ

Question

[_____]

Option 1

[_____]

Option 2

[_____]

Option 3

[_____]

Option 4

[_____]

Correct Answer

[_____]

[Save]

7. Subscription

Admin can create subscription plans.

Grid

ID	Subscription	Duration	Amount	Status	Action
----	--------------	----------	--------	--------	--------

1	Basic	1 Month	499	Active	Edit Delete
2	Premium	6 Month	1999	Active	Edit Delete

Add Modal

Subscription Name

[_____]

Duration

[_____]

Amount

[_____]

Status

[Active ▼]

[Save]

8. Admin Dashboard

Dashboard can show cards such as:

Total Users

125

Total Courses

20

Total Sub Courses

45

Total Enrollments

250

Total Revenue

₹1,25,000

9. User Side

User menu:

User Dashboard

Home

All Courses

My Courses

Subscription

Profile

Logout

10. All Courses

Courses will be displayed as Bootstrap cards.

ASP.NET Core	
Learn ASP.NET Core MVC	
Price: ₹999	
[Buy Now]	

User can open the course details and purchase it.

11. Razorpay Payment

The actual purchase flow will be:

Course



Buy Now / Add To Cart



Create Razorpay Order



Razorpay Checkout



User Pays



Payment Response



Server-side Signature Verification



Payment Successful



Create Enrollment



My Courses

We will use **Razorpay Test Mode** during development.

The Razorpay secret key will remain on the **server side**.

12. Payment

We should maintain payment information such as:

Payment

PaymentId

UserId

MasterCourseId

Amount

RazorpayOrderId

RazorpayPaymentId

RazorpaySignature

PaymentStatus

PaymentDate

After successful payment:

Payment

↓

Success

↓

Enrollment Created

↓

Course Added To My Courses

If payment verification fails:

Payment

↓

Failed

↓

No Enrollment

13. My Courses

After purchasing a course:

My Courses

ASP.NET Core

|

+-- MVC

| |

| +-- Introduction

| +-- Controllers

| +-- Views

|

+-- Entity Framework

|

+-- DbContext

+-- LINQ

User can click a topic and access its material/video.

14. User Profile

Profile will contain:

My Profile

Name

Email

Mobile

Address

Profile Image

[Edit Profile]

15. User Dashboard

Since Profile and Dashboard are different requirements, we should create both.

Dashboard can show:

Welcome, User

Total Courses

5

Completed Courses

2

Active Subscription

Premium

Learning Progress

60%

16. Database Structure

Main tables:

Users

MasterCourses

SubCourses

Topics

Materials

Subscriptions

UserSubscriptions

Payments

Enrollments

Relationships:

User

|

+----- Payment

|

+----- Enrollment

|

+----- UserSubscription

MasterCourse

|

+----- SubCourse

|

+----- Topic

|

+----- Material

Subscription

|

+----- UserSubscription

17. Main Models

Models

|

└─ User.cs

- └─ MasterCourse.cs
- └─ SubCourse.cs
- └─ Topic.cs
- └─ Material.cs
- └─ Subscription.cs
- └─ UserSubscription.cs
- └─ Payment.cs
- └─ Enrollment.cs

And:

Data

- └─ ApplicationDbContext.cs

18. Main Controllers

Controllers

- |
- └─ AccountController.cs
- |
- └─ AdminDashboardController.cs
- └─ MasterCourseController.cs
- └─ SubCourseController.cs
- └─ TopicController.cs
- └─ MaterialController.cs
- └─ SubscriptionController.cs
- |
- └─ UserDashboardController.cs
- └─ CourseController.cs
- └─ CartController.cs
- └─ PaymentController.cs
- └─ MyCourseController.cs
- └─ ProfileController.cs

19. Final Complete Flow

Admin

Login



Admin Dashboard



Master Course



Sub Course



Topic



Material



Subscription

User

Register / Login



User Dashboard



All Courses



Select Course



Buy Now



Razorpay Checkout



Payment Verification



Enrollment



My Courses



Open Course



Topic



Video / Assignment / MCQ